

Series of exercises No.3

Exercise 1:

Give the truth table of the following logical expressions:

1. $a + a.b$
2. $a + \bar{a}.b$
3. $a.b.c + b.c + c.\bar{b}.c$
4. $(a + b)(a + c)$

Exercise 2:

Prove the following property: $A + \bar{A}B = A + B$

1. with truth table.
2. Algebraically.

Exercise 3:

Give the complement of the following functions:

1. $F(A, B, C) = A + \bar{B}.C$
2. $F(A, B, C) = A.B + C$
3. $F(A, B, C) = (A.B + C)(A + B.C)$

Exercise 4 :

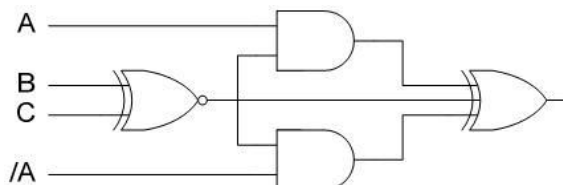
Simplify the expressions below algebraically:

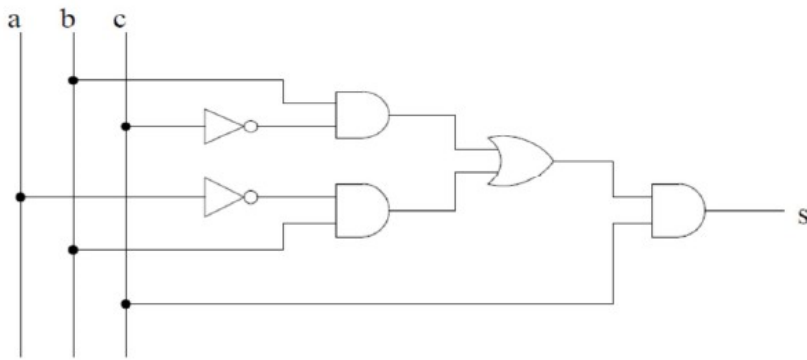
1. $\overline{A.B + A.\bar{B} + \bar{A}.\bar{B}.C}$
2. $B + A.\bar{B}.C$

Exercise 6 :

Analyze the following combinational circuits and give the output function of each circuit:

1-



**Exercise 7 :**

1. Prove the following relations algebraically:

- $(A \cdot \bar{B} + C) + (\bar{A} + B) \cdot \bar{C} = 1$
- $\overline{A \cdot C + B \cdot \bar{C}} = \bar{A} \cdot C + \bar{B} \cdot \bar{C}$

2. Express the following function in the first and second canonical form:

- $f_2(a, b, c) = 1$ if the count of variables at 1 is even (*pair*)

Exercise 8:

Either the following function: $F(A, B, C) = (A \cdot B + C)(AC + B)$

- Give the second algebraic canonical form without using the truth table (FC2).
- Deduce the first canonical decimal form (FC1).
- Deduce the first algebraic canonical form (FC1).

Exercise 9:

1. Simplify the following KARNAUGH table:

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a)

		BA			
		00	01	11	10
DC	00	1			
	01	1		X	
	11	1			X
	10	1			

b)

		CD			
		00	01	11	10
AB	00				1
	01	X	1	1	1
	11	X	1	1	X
	10	X			

c)

		BA			
		00	01	11	10
DC	00	1	1		1
	01	1	1	X	1
	11	1	1		
	10	X	X		

		cd			
		00	01	11	10
ab	00	1	0	1	1
	01	1	0	1	1
	11	0	0	0	0
	10	1	0	0	0

		cd			
		00	01	11	10
ab	00	0	1	1	0
	01	1	0	1	0
	11	0	1	0	1
	10	1	0	0	1

		cd			
		00	01	11	10
ab	00	0	1	0	0
	01	0	0	0	0
	11	0	0	0	1
	10	1	0	0	1

2. Simplify the following logical functions with KARNAUGH table:

1. $F1(A, B, C) = \sum m(0, 1, 2, 5, 6, 7)$

2. $F2(A, B, C, D) = \sum m(0, 2, 5, 6, 10, 11, 13, 14, 15)$

3. Draw the logic circuit for each simplified function using only NAND gates (NOR gates).